Name of Document	Scope of Work
Name of Project	CHIETA Data Lab — Data Science Solution & Prototype Development

1. Background

CHIETA is establishing a Data Lab — a practical environment where selected organisational datasets can be analysed, explored, and used to develop innovative solutions. The initiative is aligned with CHIETA's Fusion 2030 strategy and is aimed at building internal data capability.

The Data Lab is not a replacement for any existing CHIETA system. It is a separate, controlled environment intended to support experimentation and learning.

What makes this project unique:
The service provider will work directly with students as co-developers. The students are part of the process — not just users. The end goal is a working prototype that demonstrates real value using real CHIETA data.

2. Purpose and Objective

The service provider is required to design, build, and deliver a functional Data Science data repository — referred to throughout this document as the Data Lab — and to ensure that the following is achieved by the end of the engagement:

- A secure, solution that allows data to be uploaded, extracted, managed, analysed, visualised, and experimented with.
- At least one functional prototype that demonstrates practical use of CHIETA data.
- Students are actively involved in building the solution/data lab and the prototype.
- Full transfer of ownership to CHIETA, including all source code, documentation, configurations, and intellectual property, with clear licensing disclosures.

3. What Needs to Be Built

The Data Lab must be a single, integrated solution with the following capabilities. The solution should allow users log in, access data, do analysis, build dashboards, test ideas, and produce outputs. All CHIETA data must be brought together into a repository that enables easy access by data users, for them to analyse and manipulate independently from the originating solution.

3.1 Technical Architecture & Stack Requirements

- The provider must propose and justify a technology stack (e.g., Python/R, open-source BI/ML tools, relational/NoSQL databases) during D2.The solution/Platform must support containerisation (e.g., Docker), version control (e.g., Git), and CI/CD readiness to ensure maintainability. **(favourably Kubernetes (AKS) over Docker for containerisation, given we have Azure kubernetes is a better toolset needed for scalability and maintainability.)**
- APIs must be documented to support future integrations, though direct integration with legacy CHIETA systems is out of scope.
- All third-party tools/libraries must use permissive open-source licenses (MIT, Apache 2.0, BSD) or be pre-approved by CHIETA in writing.

3.2 Data Access and Management & Governance

Users must be able to bring data into the data lab and manage it.

- Upload datasets in common formats (e.g. Excel, CSV)
- View, search, and organise uploaded datasets.
- Clean and prepare data within the system (basic data preparation tools)
- Ensure data is stored securely and is not accessible to unauthorised users.
- All data must be encrypted at rest and in transit. Access must be logged. Sensitive fields must be masked/pseudonymised where required.
- The Lab must comply with POPIA and applicable data protection standards.

A Data Governance Addendum must be submitted with D2.

3.3 Data Analysis and Exploration

Users must be able to work with data to find patterns and generate insights.

- Run basic statistical analyses on datasets.
- Filter, sort, and explore data interactively.
- Support for both technical users (e.g. scripts/queries, SQL/Python notebooks, query editors) and non-technical users (e.g. point-and-click tools, drag-and-drop, guided workflows)
- All analytical processes must be auditable and reproducible.
- Data pipelines for processing data to and from CHIETA systems for analysis or updates to another system.
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3.4 Visualisation — Dashboards and Reports

Users must be able to see their data in visual formats that are easy to understand.

- Build charts, graphs, and visual summaries from data.
- Create and save custom dashboards.

- Export generated reports in standard formats (PDF, Excel, PNG)
- Dashboards must meet WCAG 2.1 AA accessibility guidelines for colour contrast, keyboard navigation, and screen reader compatibility.

3.5 Experimentation Tools

The system must allow users to test ideas, scenarios, and models using data.

- Ability to test hypotheses against datasets.
- Run scenario comparisons (e.g. what-if analyses)
- Support for building and testing basic predictive or analytical models.
- Ethics & Bias Review: Any model development must include a documented fairness/bias assessment and explainability summary.

3.6 User Management and Access Control

- Platform must be accessible via a web browser — no special software installations needed.
- Multiple users must be able to log in with individual accounts.
- Role-based access: different user types (admin, analyst, viewer) see different things.
- Secure individual user authentication (SSO integration preferred if feasible).
- Session timeout, password policy, and audit logging mandatory.

3.6 Working Prototype

The service provider must facilitate and deliver at least one functional, demonstrable prototype built on the Data Lab using real CHIETA data. This is the most important output of the project.

What is the prototype?

The prototype is a fully working, demonstrable solution built on top of the Data Lab using real CHIETA data. It must solve a real problem or answer a real question relevant to CHIETA operations — for example, a dashboard that tracks learner enrolment trends, a tool that identifies grant spending patterns, or a model that flags risks in a dataset.

The prototype must be usable, testable, and presentable to CHIETA leadership.

- The provider and CHIETA will agree on the prototype topic during the first phase using a weighted matrix evaluating: business impact, data availability, student learning value, technical feasibility, and leadership demonstrability. Students will be central to building it.

- Minimum Viability Requirements:
    - Solves a clearly defined CHIETA operational question.
    - Includes ≥3 interactive visualisations.
    - Supports ≥2 user roles.
    - Processes agreed dataset volume in ≤3 seconds per query/dashboard load.
    - Fully testable, documented, and presentable to CHIETA leadership.

4. Datasets

CHIETA will provide selected datasets for use in the Data Lab. The provider must not use any CHIETA data outside the agreed system environment. Initial datasets are likely to include:

- Grants data (DG and MG)
- Learner enrolment and completion data
- Sector skills and labour market data.
- Smart Skills Centre utilisation data
- ETQA data according to APP targets

The provider is responsible for preparing (cleaning and structuring) datasets once they are provided by CHIETA, and ensuring data is correctly integrated into the system:

- Conduct an initial data profiling sprint in Phase 1 to assess quality, completeness, and sensitivity.
- Clean, structure, anonymise where necessary, and load datasets into the system.
- Document all data transformations, lineage, and quality assumptions in D4.

5. Student Involvement

This is not a standard software development project.
The service provider is expected to actively involve students in the development process. Students are not observers — they are contributors. The provider must plan the engagement so that students can meaningfully participate in building both the Data Lab and the prototype.

In practical terms, this means the provider must:

1. **Structure Work for Participation:** Use agile sprints, paired programming, code reviews, and guided workshops.
2. **Define Student Roles & Time Commitment:** Assign clear roles (e.g., Data Assistant, QA Tester, Documentation Contributor, UI/UX Reviewer) with expected weekly hours (typically 8–12 hrs/student).
3. **Mentorship & Onboarding:** Deliver a Student Onboarding & Mentorship Plan in D1, covering technical training, code of conduct, confidentiality agreements, and IP acknowledgements.
4. **Supervision & Quality Control:** All student code/contributions must be reviewed, approved, and integrated by the provider's lead technical staff before deployment.
5. **Documentation & Recognition:** Maintain a contribution log. Recommend digital badges, certificates, or capstone credits to incentivise quality work.

CHIETA will facilitate access to the students and coordinate their availability. The provider is responsible for mentoring and directing their contribution.

6. Deliverables

The following must be delivered by the service provider. CHIETA will sign off on each deliverable before the next phase commences.
Note that the Memorandum of Agreement (MOA) sets out the full list of documentation and deliverables that must be submitted as a condition for payment to be released — the table below provides a summary for reference.

#	Deliverable	Description	Sign-off
D1	**Requirements Specification Document**	Stakeholder-aligned functional/non-functional requirements; prototype topic selection matrix; student onboarding & mentorship plan.	CHIETA approval required
D2	**System Design Document**	Technical architecture, data flows, stack justification, API map, role-permission matrix, Data Handling & Security Protocol (POPIA compliance).	CHIETA approval required
D3	**Functional Data Lab**	Fully deployed, browser-accessible solution with all	User Acceptance Testing

#	Deliverable	Description	Sign-off
		Section 3 capabilities. Environment setup documented.	
D4	**Integrated and Prepared Datasets**	Cleaned, validated, anonymised (if required), and loaded datasets with data lineage & quality report.	Data verification by CHIETA
D5	**Dashboards and Reports**	≥2 fully functional dashboards with export capabilities, accessibility compliance, and user guides.	CHIETA approval
D6	**Working Prototype**	Leadership-ready prototype meeting minimum viability criteria, tested with real data, and accompanied by a business value summary.	Formal sign-off by CHIETA
D7	**Testing and Validation Report**	Test strategy, unit/integration/UAT scripts, performance benchmarks, accessibility check, security scan results, and defect resolution log.	CHIETA sign-off
D8	**Training and User Guides**	Role-based user manuals, video tutorials, train-the-trainer session delivery, and a Maintenance & Troubleshooting Handbook.	Completion confirmation
D9	**Source Code and Technical Documentation**	Complete repository, deployment instructions, dependency/license audit, CI/CD pipeline docs, and architecture diagrams.	CHIETA ownership confirmed

7. Phases and Suggested Timeline

The project must be completed in four phases. Timelines below are indicative — the provider must confirm a detailed project plan within two weeks of appointment.

#	Phase	Key Activities	Timeline
1	**Business Requirements & Design**	Stakeholder engagement, requirements gathering, system design, agree on prototype topic, project plan	2 Months
2	**Development**	Core system build, dataset integration, dashboard development, student sprint cycles, iterative feedback, version control implementation	8 Months
3	**Prototype & Testing**	Prototype refinement, functional/UAT execution, performance & accessibility testing, security validation, defect resolution	4 Months
4	**Training & Handover**	Train-the-trainer sessions, documentation finalisation, source code transfer, environment handover, formal sign-off	2 Months

Total estimated duration: 18 Months as per MOA.

8. Assumptions

This scope of work is based on the following assumptions. If any of these do not hold, the scope or timeline may need to be revised.

- CHIETA will provide the required datasets in a usable format within the agreed timeframe.
- CHIETA will make relevant stakeholders available for the requirements gathering sessions.
- Students will be made available for scheduled development and workshop sessions — the provider will be informed in advance of any changes to availability.
- The service provider has full authority and responsibility for technical decisions related to the system design and build, subject to CHIETA approval of D2 and D9 licensing disclosures.

- The Data Lab will be hosted in an environment agreed between CHIETA and the provider — hosting costs are not included in this scope unless otherwise agreed.
- Infrastructure provisioning, backups, and monitoring are CHIETA responsibilities unless otherwise contracted.

- All CHIETA data will be handled in strict compliance with POPIA and organisational data policies.
- The prototype topic will be locked after Phase 1 sign-off. Changes thereafter require formal change control (see Appendix B).
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- The provider will use tools and technologies that are appropriate for the project and which CHIETA can practically maintain after handover & will be responsible for the costs associated during the development phase including tooling & licensing costs.

9. What Is Not Included

To avoid any confusion, the following are not part of this scope of work:

- Replacement or integration with CHIETA's existing operational systems (e.g. grants management system, HR systems)
- Ongoing support or maintenance after the handover — unless a separate agreement is made.
- Licensing costs for commercial software tools, cloud hosting, or third party APIs, unless agreed in advance formally.
- Hardware procurement or on-premise server management.
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10. What Success Looks Like

At the end of this project, CHIETA should be able to:

- Log into a secure, functioning Data Lab and access approved datasets.
- Run analyses, build dashboards, and generate reports from within the system independently.
- Demonstrate at least one working prototype to leadership that solves a real problem.
- Hand the system to a developer or administrator and have them understand how it works.
- Point to clear student contributions throughout the build.
- Hand the system to an internal administrator/developer with clear documentation and training.

- Demonstrate measurable student contributions across development sprints.

11. Quantitative Success Indicators:

- System uptime ≥ 99% during UAT and handover period.
- Dashboard/data query load time ≤ 3 seconds for datasets up to 500k rows.
- ≥ 80% of trained CHIETA staff report ≥ 4/5 confidence in independent system usage.
- 100% of student contributions documented, reviewed, and version controlled.
- Zero critical/high severity defects remaining at final sign-off.
- Full license audit completed; no restrictive (e.g., GPL, commercial) dependencies without written CHIETA approval.

The single most important outcome of this project is the working prototype.
Everything else supports getting there. If the prototype is compelling, usable, and grounded in real CHIETA data, this project will be a success.

11. Sign-Off

By accepting this Scope of Work, the service provider confirms that they have read, understood, and agree to deliver the requirements described in this document.

On behalf of CHIETA	On behalf of Service Provider
Name: _____	Name: _____
Title: _____	Title: _____
Sign: _____	Sign: _____
Date: _____	Date: _____

APPENDIX A: RISK REGISTER & MITIGATION

Risk	Impact	Likelihood	Mitigation Strategy
Student availability/turnover	Delays in sprint delivery	Medium	Modular task design; cross-training; 15% schedule buffer; CHIETA backup pool
Poor data quality in source datasets	Rework in D3/D4	High	Phase 1 data profiling sprint; data cleansing buffer; clear quality thresholds

Risk	Impact	Likelihood	Mitigation Strategy
Prototype scope creep	Timeline overrun, budget pressure	Medium	Locked prototype criteria post-Phase 1; formal change control process
Technology stack not maintainable by CHIETA	Post-handover sustainability risk	Low-Medium	D2 maintainability assessment; prefer open-source, widely documented tools
Insufficient user adoption	Low ROI	Medium	Early stakeholder involvement in design; iterative UAT; train-the-trainer model
POPIA/data breach non-compliance	Legal/reputational damage	Low	D2 Data Handling Protocol; encryption, masking, audit logs; security testing in D7

APPENDIX B: CHANGE CONTROL PROCESS

1. **Request:** Either party submits a Change Request Form detailing scope, timeline, cost, and risk impact.
2. **Review:** CHIETA Project Sponsor and Provider Lead assess within 5 working days.
3. **Approval:** Changes require written sign-off from both parties before implementation.
4. **Documentation:** Approved changes update D1/D2, project plan, and deliverable baselines.
5. **Expedited Changes:** Critical security/compliance changes may be implemented immediately with retrospective documentation.

APPENDIX C: GLOSSARY

Term	Definition
Data Lab	Secure, browser-based environment for data access, analysis, visualisation, and experimentation.
Prototype	A fully functional, demonstrable solution built on the Data Lab using real CHIETA data to address a specific business question.
POPIA	Protection of Personal Information Act (South Africa), governing lawful data processing and security.
WCAG 2.1 AA	Web Content Accessibility Guidelines ensuring digital tools are usable by people with disabilities.
CI/CD	Continuous Integration/Continuous Deployment pipeline for automated testing and deployment.

Term	Definition
UAT	User Acceptance Testing — validation by CHIETA stakeholders that the system meets business requirements.